

D A circuit is needed to realize the impulse response transform shown below. Scale the circuit so that all parts use practical values.

$$H(s) = \pm \frac{200s + 10^6}{s^2 + 200s + 10^6}$$

E It is claimed that both circuits in Figure P11-58 realize the transfer function

$$T_V(s) = K \left(\frac{s + 2000}{s + 1000} \right)$$

Verify that both circuits realize the specified $T_V(s)$. Which circuit would you choose if the output must drive a $1\text{ k}\Omega$ load?

Which circuit would you choose if the input comes from a $10\text{-}\Omega$ source?

It is further claimed that connecting the two circuits in cascade produces an overall transfer function of $[T_V(s)]^2$ no matter which circuit is the first stage and which is the second stage.

Do you agree or disagree? Explain.

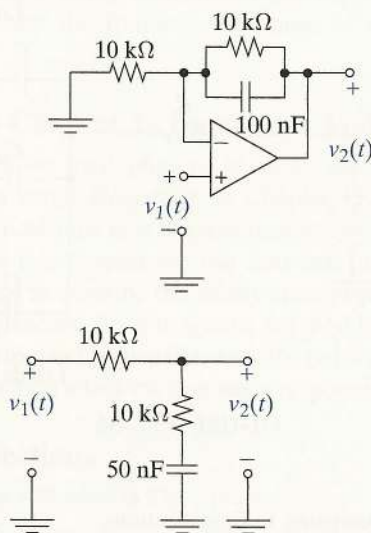


FIGURE P11-58

E It is claimed that both circuits in Figure P11-59 realize the transfer function

$$T_V(s) = \frac{\pm 1000s}{(s + 1000)(s + 4000)}$$

Verify that both circuits realize the specified $T_V(s)$. Which circuit would you choose if the output must drive a $1\text{ k}\Omega$ load?

Which circuit would you choose if the input comes from a $10\text{-}\Omega$ source?

It is further claimed that connecting the two circuits in cascade produces an overall transfer function of $[T_V(s)]^2$ no

matter which circuit is the first stage and which is the second stage.

Do you agree or disagree? Explain.

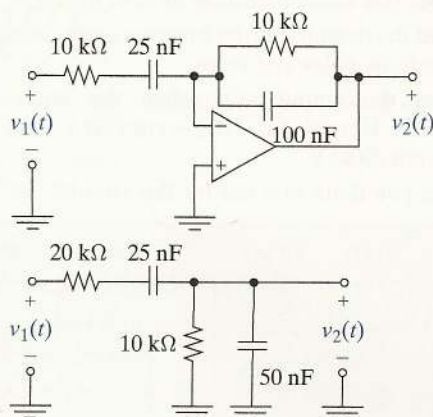


FIGURE P11-59

11-60 D Design a circuit that produces the following step response:

$$g(t) = 10[1 - e^{-50t} - 50te^{-50t}]u(t)$$

11-61 D A circuit is needed that will take an input of $v_1(t) = 2e^{-100t}u(t)$ V and produce an output of $v_2(t) = 5e^{-1000t}u(t)$ V. Design such a circuit using practical part values. Validate your design by using OrCAD.

11-62 D A circuit is needed that will take an input of $v_1(t) = [1 - e^{-10,000t}]u(t)$ V and produce a constant -5 V output. Design such a circuit using practical part values. Validate your design using OrCAD.

INTEGRATING PROBLEMS

11-63 A D First-Order Circuit Impulse and Step Responses Each row in the table shown in Figure P11-63 refers to a first-order circuit with an impulse response $h(t)$ and a step response $g(t)$. Fill in the missing entries in the table.

Circuit	$h(t)$	$g(t)$
	$\delta(t) - [\alpha e^{-\alpha t}]u(t)$	
		$\left(1 + \frac{e^{-\alpha t}}{2}\right)u(t)$

FIGURE P11-63

INTEGRATING PROBLEMS

A Thévenin's Theorem from Time-Domain Data
A black box containing a linear circuit has an on/off switch
a pair of external terminals. When the switch is turned
the open-circuit voltage between the external terminals
observed to be

$$v_{OC}(t) = (18e^{-10t} - 12e^{-50t})u(t) \text{ V}$$

the short circuit current was observed to be

$$i_{SC}(t) = [138.5e^{-10t} - 174.3e^{-50t} + 32.9e^{-96.6t}]u(t) \text{ mA}$$

A load resistance is connected across the terminals and
the switch turned on again. What is the voltage delivered to
the load?

Design a Load Impedance

Consider to match the Thévenin impedance of a source, the
load impedance in Figure P10-53 must be

$$Z_L(s) = \frac{s+5}{s+10}$$

What impedance $Z_2(s)$ is required?

How would you realize $Z_2(s)$ using only resistors, in-
ductors, and/or capacitors? (Hint: Write $Z_L(s)$ as a sum of
partial fractions, then solve for $Y_2(s)$.)

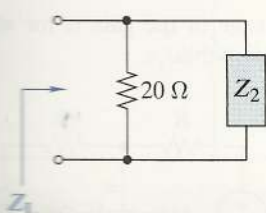


FIGURE P10-53

RC Circuit Analysis and Design

The RC circuits in Figure P10-54 represent the situation
of the input to an oscilloscope. The parallel combination
of R_1 and C_1 represents the probe used to connect the
oscilloscope to a test point. The parallel combination of
 R_2 and C_2 represents the input impedance of the
oscilloscope.

Assuming zero initial conditions, transform the
circuit into the s domain and find the relationship between
the test point voltage $V_S(s)$ and the voltage $V_O(s)$ at the
oscilloscope's input.

For $R_2 = 10 \text{ M}\Omega$ and $C_2 = 2 \text{ pF}$, determine the
values of R_1 and C_1 that make the input voltage a scaled
duplicate of the test point voltage.

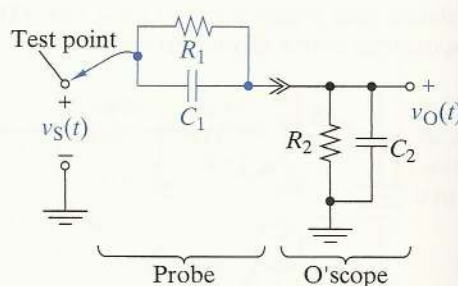


FIGURE P10-54

10-55 s-Domain OP AMP Circuit Analysis

The OP AMP circuit in Figure P10-55 is in the zero state.
Transform the circuit into the s domain and use the OP AMP
circuit analysis techniques developed in Sect. 4-4 to find the
relationship between the input $V_1(s)$ and the output $V_2(s)$.

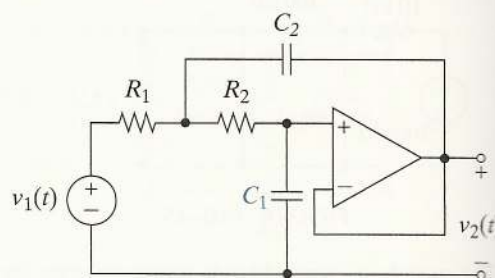


FIGURE P10-55

10-56 Pulse Conversion Circuit

The purpose of the test setup in Figure P10-56 is to deliver
damped sine pulses to the test load. The excitation comes
from a 1-Hz square wave generator. The pulse conversion
circuit must deliver damped sine waveforms with $\zeta < 0.5$ and
 $\omega_0 > 10 \text{ krad/s}$ to $50\text{-}\Omega$ and $600\text{-}\Omega$ loads. The recommended
values for the pulse conversion circuit are $L = 10 \text{ mH}$ and
 $C = 100 \text{ nF}$. Verify that the test setup meets the specifications.
(Hint: Compute the voltage across the load for an input
signal equal to a unit step function, $u(t)$, and then again for a
negative unit step function, $-u(t)$. Note that the output of the
square wave generator is the sum of a series of step functions.)

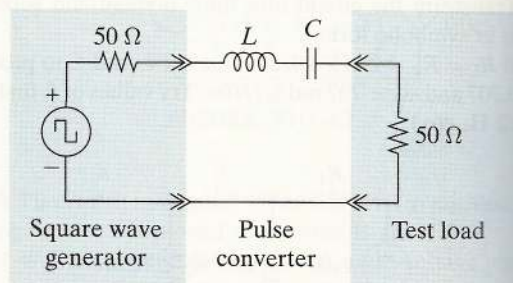


FIGURE P10-56

NETWO

The driving-point impedance
the resulting current at the

Some History Behi

The network function co
the development of syste
filters for long-distance t
effort eventually evolue
synthesis. The purpose
circuits that produce a
Foster along with Sid
Wilhelm Cauer, and Ott
the founders of modern

Why This Chapter

This special chapter intro
concepts of electrical engi
this chapter you will learn
why they are important de
importantly, you will learn
realize a desired network
Since design can lead to
face you to the criteria use

Chapter Sections

- 10-1 Definition of a Network Function
- 10-2 Network Functions of Resistive Circuits
- 10-3 Network Functions of RC and RL Circuits
- 10-4 Network Functions of RLC Circuits
- 10-5 Network Functions of Two-Port Networks
- 10-6 Impulse Response and Step Response
- 10-7 Network Function Determination

zero-state response $y(t)$. Symbolically the convolution integral is represented by $y(t) = h(t) * x(t)$. Time-domain convolution and s -domain multiplication are equivalent; that is, $y(t) = h(t) * x(t) = \mathcal{L}^{-1}\{H(s)X(s)\}$. The geometric interpretation of t -domain convolution involves four operations: reflecting, shifting, multiplying, and integrating.

- First- and second-order transfer functions can be designed using voltage dividers and inverting or noninverting OP AMP circuits. Higher-order transfer functions can be realized using a cascade connection of first- and second-order circuits. Prototype designs usually require magnitude scaling to obtain practical element values.

PROBLEMS

OBJECTIVE 11-1 NETWORK FUNCTIONS (SECTS. 11-1, 11-2)

- Given a linear circuit, find specified network functions and locate their poles and zeros.
- Given a linear circuit, select the element values to produce specified poles and zeros. See Examples 11-2, 11-3, 11-4, 11-5, 11-6, 11-7 and Exercises 11-2, 11-3, 11-4, 11-5, 11-6, 11-7

11-1 Find the driving-point impedance seen by the voltage source in Figure P11-1 and the voltage transfer function $T_V(s) = V_2(s)/V_1(s)$.

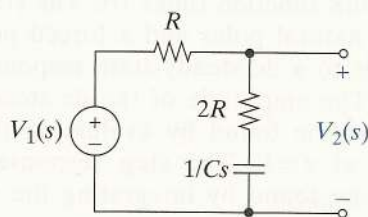


FIGURE P11-1

11-2 Find the driving-point impedance seen by the voltage source in Figure P11-2 and the voltage transfer function $T_V(s) = V_2(s)/V_1(s)$.

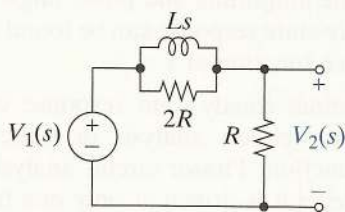


FIGURE P11-2

11-3 Find the driving-point impedance seen by the voltage source in Figure P11-3 and the voltage transfer function $T_V(s) = V_2(s)/V_1(s)$.

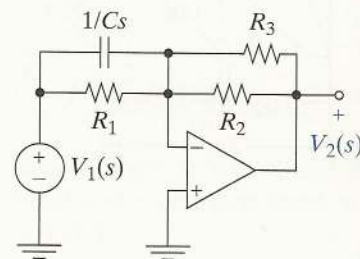


FIGURE P11-3

11-4 Find the driving-point impedance seen by the voltage source in Figure P11-4 and the voltage transfer function $T_V(s) = V_2(s)/V_1(s)$.

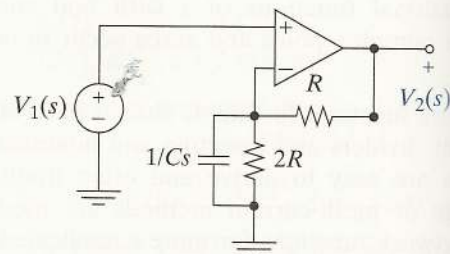


FIGURE P11-4

11-5 Find the voltage transfer function $T_V(s) = V_2(s)/V_1(s)$ for the circuit in Figure P11-5.

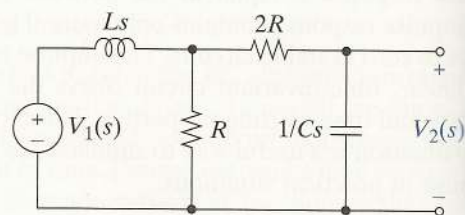
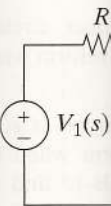


FIGURE P11-5

Find the driving-point impedance seen by the voltage source in Figure P11-6 and the voltage transfer function $T_V(s) = V_2(s)/V_1(s)$.



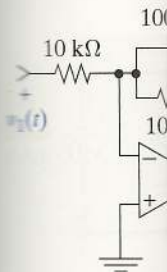
Find the driving-point impedance seen by the voltage source in Figure P11-7 and the voltage transfer function $T_V(s) = V_2(s)/V_1(s)$.



Find the current $I_1(s)$ in Figure P11-8. Select the value of R such that the current $I_1(s)$ is zero at $s = -500$ rad/s.



Find the voltage transfer function $T_V(s) = V_2(s)/V_1(s)$ for the cascade connection of the two circuits in Figure P11-9 and locate the poles and zeros of the transfer function.



7 The circuit in Figure P11-27 is in the steady state with $v_1(t) = 10 \cos 2000t$ V. Find $v_{2ss}(t)$. Repeat for $v_1(t) = 10 \cos 10kt$ V.

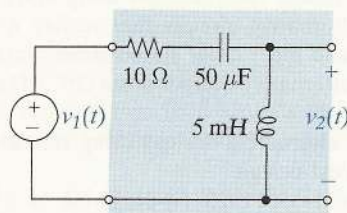


FIGURE P11-27

8 The circuit in Figure P11-28 is in the steady state with $i_1(t) = 10 \cos 500t$ mA, $R_1 = 100 \Omega$, $R_2 = 400 \Omega$, and $L = 100$ mH. Find $i_{2ss}(t)$. Repeat for $i_1(t) = 10 \cos 5000t$ mA.

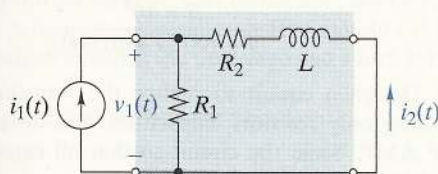


FIGURE P11-28

9 The OrCAD circuit drawn in Figure P11-29 is in the steady state with $i_1(t) = 5 \cos 1000t$ mA, $R = 1$ k Ω , $L = 2$ H, and $C = 500$ nF. Use OrCAD to find $i_{2ss}(t)$. Copy the figure in OrCAD. Draw your circuit and use IAC for a source. The PRINT element on the right side of the figure is an ammeter and it comes from the Special Library. It has a polarity as indicated by the small “-” on the element. You should ensure that it will read the correct current direction. Before you can use the PRINT element, it must be set up to display the current magnitude and phase. Double-click on the element and in its Property Editor place a “y” in the AC, MAG, and PHASE boxes. Click the Apply button and close the Property Editor. To obtain the simulation, select AC Sweep/Noise. Set the start frequency at 159 Hz (1000 rad/s) and the stop frequency at the same value. Place a 1 in Points/Decade. Run the simulation. Under View, select Output File. Scroll down until you see the PRINT output. Repeat for $i_1(t) = 5 \cos 10kt$ mA.

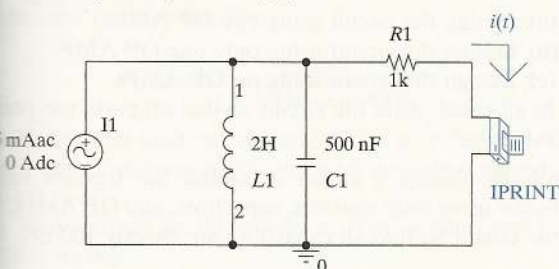


FIGURE P11-29

11-30 The circuit in Figure P11-30 is in the steady state with $i_1(t) = 10 \cos 5000t$ mA. Find the steady-state voltage $v_{2ss}(t)$. Repeat for $i_1(t) = 5 \cos 2500t$ mA.

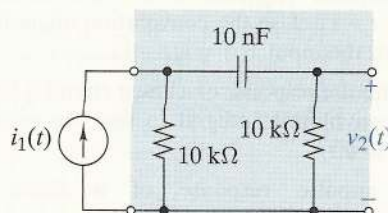


FIGURE P11-30

11-31 The transfer function of a linear circuit is $T(s) = (s + 500)/(s + 50)$. Find the sinusoidal steady-state output for the input $x(t) = 5 \cos 500t$.

11-32 The step response of a linear circuit is $g(t) = [5e^{-500t}]u(t)$. Find the sinusoidal steady-state output for the input $x(t) = 2 \cos 1000t$.

11-33 The step response of a linear circuit is $g(t) = [2e^{+100t}]u(t)$. Find the sinusoidal steady-state output for the input $x(t) = 5 \cos 500t$.

11-34 The impulse response of a linear circuit is $h(t) = [800e^{-1000t}]u(t) - \delta(t)$. Find the sinusoidal steady-state output for the input $x(t) = 3 \cos 400t$.

11-35 The impulse response of a linear circuit is $h(t) = 400[e^{-100t} - e^{-400t}]u(t)$. Use MATLAB to find the sinusoidal steady-state output for the input $x(t) = 5 \cos 200t$. Use MATLAB to plot $y(t)$.

11-36 The step response of a linear circuit is $g(t) = [e^{-60t} \sin 80t]u(t)$. Find the sinusoidal steady-state response for the input $x(t) = 20 \cos 100t$.

11-37 The step response of a linear circuit is $g(t) = [1 - 20te^{-10t}]u(t)$. Find the sinusoidal steady-state response for the input $x(t) = 50 \cos 10t$.

OBJECTIVE 11-4 NETWORK FUNCTIONS AND CONVOLUTION (SECT. 11-6)

(a) Given the impulse response of a linear circuit, use the convolution integral to find the response to a specified input.

(b) Use the convolution integral to derive properties of linear circuits.

See Examples 11-15, 11-16, 11-17, 11-18 and Exercises 11-17, 11-18, 11-19, 11-20

11-38 The impulse response of a linear circuit is $h(t) = [u(t) - u(t - 2)]$. Use the convolution integral to find the response to the input $x(t) = u(t - 2)$.

OBJECTIVE 11-5 NETWORK FUNCTION DESIGN (SECT. 11-7)

- Visualize design specifications using MATLAB.
- Design alternative circuits that realize a given network function and meet other stated constraints.
- Validate alternative designs using OrCAD simulation software.
- Evaluate alternative designs using stated criteria and select the best design.

See Examples 11-19, 11-20, 11-21, 11-22, 11-23, 11-24, 11-25 and Exercises 11-21, 11-22, 11-23, 11-24, 11-25.

- 11-51 **D** Design a circuit to realize the transfer function below using only resistors, capacitors, and OP AMPs. Scale the circuit so that all resistors are greater than 10 kΩ and all capacitors are less than 1 μF.

$$T_V(s) = \pm \frac{2 \times 10^5}{(s + 100)(s + 10,000)}$$

- 11-52 **D** Design a circuit to realize the transfer function below using only resistors, capacitors, and no more than one OP AMP. Scale the circuit so that all capacitors are exactly 10 nF.

$$T_V(s) = \pm \frac{100(s + 1000)}{(s + 100)(s + 10,000)}$$

- 11-53 **D** Design a circuit to realize the transfer function below using only resistors, capacitors, and no more than one OP AMP. Scale the circuit so that the final design uses only 20-kΩ resistors.

$$T_V(s) = \pm \frac{10,000s}{(s + 1000)(s + 5000)}$$

- 11-54 **D** Design a *passive* circuit to realize the transfer function below using only resistors, capacitors, and inductors. Scale the circuit so that all inductors are 50 mH or less.

$$T_V(s) = \frac{s^2}{(s + 2000)^2}$$

- 11-55 **D** A circuit is needed to realize the transfer function

$$T_V(s) = \pm \frac{(s + 100)(s + 500)}{(s + 200)(s + 1000)}$$

- Design the circuit using two OP AMPs.
 - Design the circuit using only one OP AMP.
 - Design the circuit using no OP AMPs.
- In all cases, scale the circuit so that all parts use practical values.

- 11-56 **D** Design a circuit to realize the transfer function below using only resistors, capacitors, and OP AMPs. Scale the circuit so that all capacitors are exactly 100 nF.

$$T_V(s) = -\frac{400(s + 100)}{s(s + 10,000)}$$

- 11-57 **D** A circuit is needed to realize the impulse response transform shown below. Scale the circuit so that all parts use practical values.

$$H(s) = \pm \frac{200s + 10^6}{s^2 + 200s + 10^6}$$

- 11-58 **E** It is claimed that both circuits in Figure P11-58 realize the transfer function

$$T_V(s) = K \left(\frac{s + 2000}{s + 1000} \right)$$

- Verify that both circuits realize the specified $T_V(s)$.
- Which circuit would you choose if the output must drive a 1-kΩ load?
- Which circuit would you choose if the input comes from a 50-Ω source?
- It is further claimed that connecting the two circuits in cascade produces an overall transfer function of $[T_V(s)]^2$ no matter which circuit is the first stage and which is the second stage.

Do you agree or disagree? Explain.

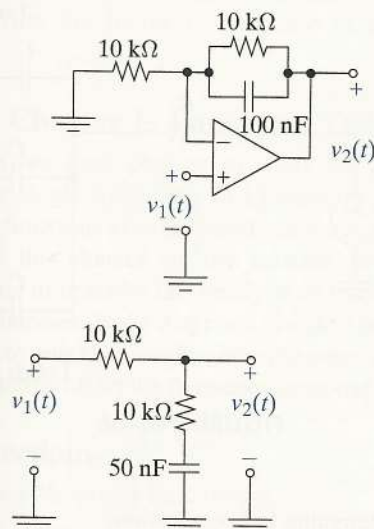


FIGURE P11-58

- 11-59 **E** It is claimed that both circuits in Figure P11-59 realize the transfer function

$$T_V(s) = \frac{\pm 1000s}{(s + 1000)(s + 4000)}$$

- Verify that both circuits realize the specified $T_V(s)$.
- Which circuit would you choose if the output must drive a 1-kΩ load?
- Which circuit would you choose if the input comes from a 50-Ω source?
- It is further claimed that connecting the two circuits in cascade produces an overall transfer function of $[T_V(s)]^2$ no

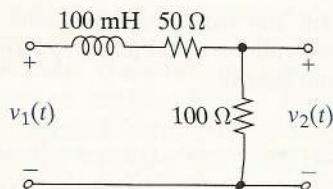


FIGURE P12-1

12-2 Find the transfer function $T_V(s) = V_2(s)/V_1(s)$ of the circuit in Figure P12-2.

- Find the dc gain, infinite frequency gain, and cutoff frequency. Identify the type of gain response.
- Sketch the straight-line approximation of the gain response.
- Calculate the gain at $\omega = 0.5\omega_C$, ω_C , and $2\omega_C$.
- Use OrCAD to plot the Bode magnitude gain response of the circuit. Validate your answers for part (c).

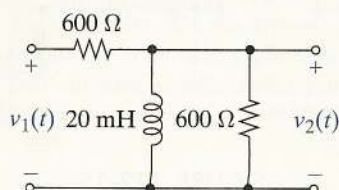


FIGURE P12-2

12-3 Find the transfer function $T_V(s) = V_2(s)/V_1(s)$ of the circuit in Figure P12-3.

- Find the dc gain, infinite frequency gain, and cutoff frequency. Identify the type of gain response.
- Sketch the straight-line approximation of the gain response.
- Calculate the gain at $\omega = 0.5\omega_C$, ω_C , and $2\omega_C$.
- Use OrCAD to plot the Bode magnitude gain response of the circuit. Validate your answers for part (c).
- What element values would you change to increase the passband gain to 10?

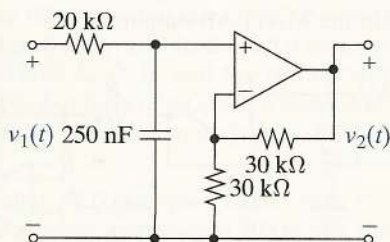


FIGURE P12-3

12-4 Find the transfer function $T_V(s) = V_2(s)/V_1(s)$ of the circuit in Figure P12-4.

- Find the dc gain, infinite frequency gain, and cutoff frequency. Identify the type of gain response.

(b) Sketch the straight-line approximation of the gain response.

(c) Calculate the gain at $\omega = 0.5\omega_C$, ω_C , and $2\omega_C$.

(d) Use OrCAD to plot the Bode magnitude gain response of the circuit. Validate your answers for part (c).

(e) What element values would you change to double the passband gain without changing the cutoff frequency?

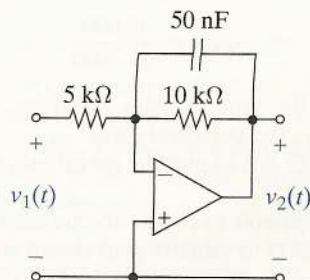


FIGURE P12-4

12-5 Find the transfer function $T_V(s) = V_2(s)/V_1(s)$ of the circuit in Figure P12-5.

- Find the dc gain, infinite frequency gain, and cutoff frequency. Identify the type of gain response.
- Use OrCAD to plot the Bode magnitude gain response of the circuit.
- What element value would you change to increase the cutoff frequency by one decade?

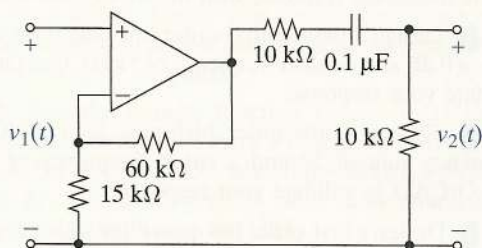


FIGURE P12-5

12-6 Find the transfer function $T_V(s) = V_2(s)/V_1(s)$ of the circuit in Figure P12-6. What type of gain response does the circuit have? What is the passband gain? Select values of R and C so that the cutoff frequency is 10 kHz.

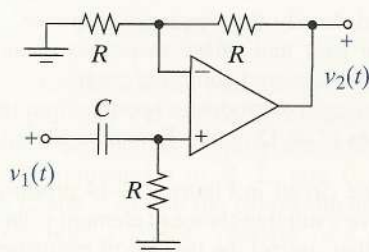


FIGURE P12-6

first-order high-pass circuit has a passband gain of 1 and a cutoff frequency of 500 rad/s. Find the gain at $\omega = 0, 200, 500$, and 1000 rad/s.

first-order low-pass circuit has a passband gain of 5 dB and a cutoff frequency of 2 krad/s. Find the gain (in dB) at 20 rad/s, 20 krad/s, and 200 krad/s.

The transfer function of a first-order circuit is

$$T(s) = \frac{10,000}{s + 1000}$$

Identify the type of gain response. Find the cutoff frequency and the passband gain.

Use MATLAB to plot the magnitude of the Bode gain.

Design a circuit to realize the transfer function.

Use OrCAD to validate your circuit design by comparing the frequency response with the MATLAB output.

The transfer function of a first-order circuit is

$$T(s) = \frac{s}{s + 3000}$$

Identify the type of gain response. Find the cutoff frequency and the passband gain.

Use MATLAB to plot the magnitude of the Bode gain.

Design a circuit to realize the transfer function.

Use OrCAD to validate your circuit design by comparing the frequency response with the MATLAB output.

Design a passive first-order low-pass filter with a dc gain of 10 dB and a cutoff frequency of 5 kHz. Use OrCAD to validate your response.

Design a first-order high-pass filter with a high-frequency gain of 50 and a cutoff frequency of 500 Hz. Use OrCAD to validate your response.

Design a first-order low-pass filter with a dc gain of 2 and a cutoff frequency of -3 dB at 1000 rad/s. Use OrCAD to validate your response.

EXERCISES 12-2 BANDPASS AND BANDSTOP FILTERS (SECT. 12-4)

Design a cascade or parallel connection of two first-order

filters and classify the frequency response.

Find the gain and phase responses using straight-line approximations and computer tools.

Design circuits to produce a specified frequency response. See Examples 12-8, 12-9 and Exercises 12-9, 12-10, 12-11

The circuit in Figure P12-14 produces a bandpass response for a suitable choice of element values. Identify the elements that control the two cutoff frequencies. Select the element values so that the passband gain is 30 and the cutoff

frequencies are 100 rad/s and 2500 rad/s. Use practical element values with $R \geq 10$ k Ω and $C \leq 1$ μ F. Use OrCAD to validate your design.

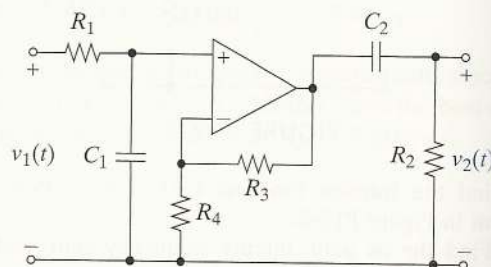


FIGURE P12-14

- 12-15 **D** Repeat Problem 12-14 for the circuit in Figure P12-15. Use OrCAD to validate your design.

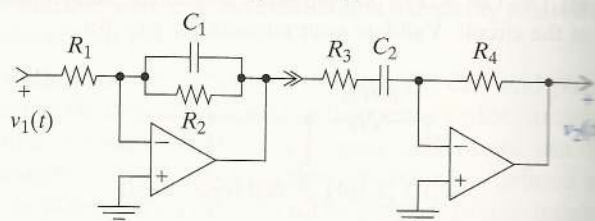


FIGURE P12-15

- 12-16 **E** Suppose the circuits you designed in Problems 12-14 and 12-15 had to feed a 500- Ω recorder. Which circuit would you select and why?

- 12-17 **D** The circuit in Figure P12-17 produces a bandpass response for a suitable choice of element values.

(a) Find the circuit's transfer function.

(b) Identify the elements that control the two cutoff frequencies. Select the element values so that the cutoff frequencies are 100 krad/s and 1000 krad/s. Use practical element values with $R \geq 10$ k Ω , $L \leq 100$ mH, and $C \leq 1$ μ F.

(c) Use MATLAB to plot the Bode magnitude plot for the values you selected.

(d) What is the passband gain for your design?

(e) Simulate your circuit using OrCAD and compare the results with the MATLAB output.

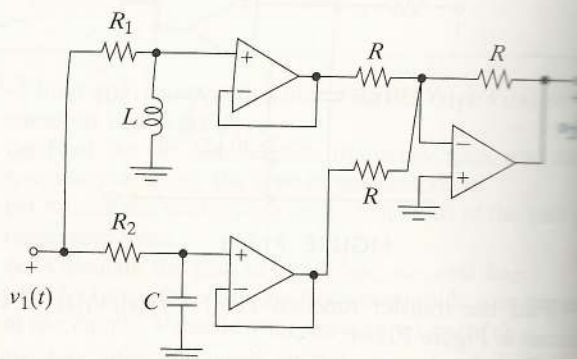


FIGURE P12-17

- 12-18 **D** The following problem involves the combination of two first-order passive filters and an inverting OP AMP. The input is a sinusoid and the output of the first filter is a bandpass circuit with the transfer function $f_{CL} = 20$ Hz, $f_{CH} = 20$ kHz, and 20 dB. Use OrCAD to validate your design.

- 12-19 **D** The following problem involves the combination of two first-order passive filters and a 50- Ω series resistor. Design a bandpass circuit with the transfer function $f_{CL} = 10$ kHz, $f_{CH} = 10$ kHz, and 20 dB. Use OrCAD to validate your design. Specifications, all $\pm 5\%$: band gain 20 dB, $B = 19$ dB. Compare your results.

- 12-20 **D** The following problem involves the combination of two first-order passive filters and an inverting OP AMP. The input is a sinusoid and the output of the first filter is a bandpass circuit with the transfer function $f_{CL} = 2$ kHz, $f_{CH} = 10$ kHz, and 20 dB. Use OrCAD to validate your design. Specifications, all $\pm 5\%$: band gain 20 dB, $B = 19$ dB. Compare your results. Why or why not?

OBJECTIVE 12-3 THE RLC CIRCUITS (SECT. 12-5)

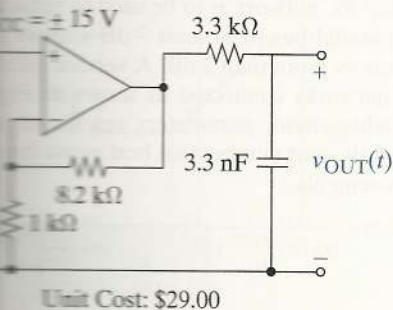
Given an RLC circuit connected to a load, find the filter response.

- (a) Find the frequency response of the filter and B .
(b) Design circuits to produce a specified frequency response.
(c) Compare the bandpass and low-pass responses from RLC filters with simple first-order filters.

See Examples 12-10, 12-11, 12-12, 12-13, 12-14, 12-15, 12-16, 12-17, 12-18, 12-19, 12-20, 12-21, 12-22, 12-23, 12-24, 12-25, 12-26, 12-27, 12-28, 12-29, 12-30, 12-31, 12-32, 12-33, 12-34, 12-35, 12-36, 12-37, 12-38, 12-39, 12-40, 12-41, 12-42, 12-43, 12-44, 12-45, 12-46, 12-47, 12-48, 12-49, 12-50, 12-51, 12-52, 12-53, 12-54, 12-55, 12-56, 12-57, 12-58, 12-59, 12-60, 12-61, 12-62, 12-63, 12-64, 12-65, 12-66, 12-67, 12-68, 12-69, 12-70, 12-71, 12-72, 12-73, 12-74, 12-75, 12-76, 12-77, 12-78, 12-79, 12-80, 12-81, 12-82, 12-83, 12-84, 12-85, 12-86, 12-87, 12-88, 12-89, 12-90, 12-91, 12-92, 12-93, 12-94, 12-95, 12-96, 12-97, 12-98, 12-99, 12-100, 12-101, 12-102, 12-103, 12-104, 12-105, 12-106, 12-107, 12-108, 12-109, 12-110, 12-111, 12-112, 12-113, 12-114, 12-115, 12-116, 12-117, 12-118, 12-119, 12-120, 12-121, 12-122, 12-123, 12-124, 12-125, 12-126, 12-127, 12-128, 12-129, 12-130, 12-131, 12-132, 12-133, 12-134, 12-135, 12-136, 12-137, 12-138, 12-139, 12-140, 12-141, 12-142, 12-143, 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project manager asks you evaluate the designs of a vendor. Which vendor would you recommend?

First-Order Filter Company



Simply Filters, Ltd

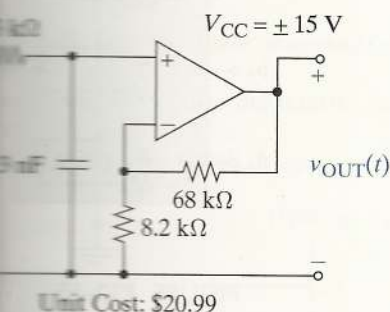


FIGURE P12-56

Evaluation
laboratory, you need a bandpass filter to meet requirements:

Design requirements: Passband gain: $10 \pm 5\%$, $B = 10 \text{ krad/s} \pm 5\%$, $\omega_0 = 5 \text{ krad/s} \pm 2\%$, $\omega_{CL} = 2 \text{ krad/s} \pm 10\%$.

Evaluation criteria: Filter performance, parts cost, use of standard parts, and cost. The two vendors have responded with the designs shown in Figure P12-57. As a research assistant, your supervisor asks you to evaluate the two designs and recommend a vendor. What vendor would you recommend and why?

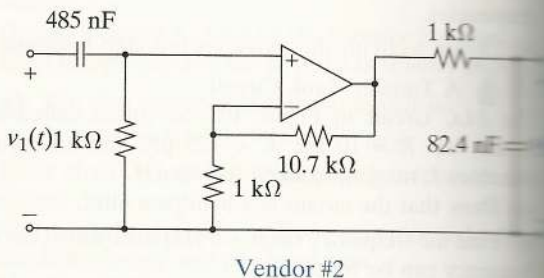
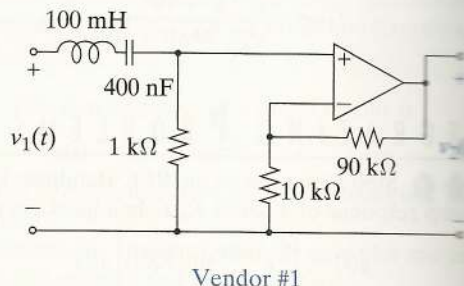


FIGURE P12-57

FOURIER ANALYSIS

The series formed of sines or cosines of different frequencies and the ordinates of lines or surfaces with respect to the frequency axis.

Some History Behind This Chapter

The analysis techniques in this chapter are based on the works of the French physicist J. B. J. Fourier. In his studies Fourier found that dissimilar periodic functions could be represented by an infinite series of sines and cosines. Fourier was guided by his intuition and presented empirical evidence to support his theory. He left the question of series convergence open. This question was later settled by the German mathematician G. L. Dirichlet (1805–1859). Fourier's theory was eventually accepted and is now widely used in many branches of science and engineering.

Why This Chapter Is Important

The word *spectrum* denotes both a natural phenomenon and a mathematical concept. As a natural phenomenon, it is used in modern communications and its use is widespread in government and international agencies. As a mathematical concept, it describes the range of frequencies needed to represent a signal. Perfect reproduction is not always necessary. Fourier theory suggests that a signal can be represented by an infinite spectrum, a practical theory also shows that a finite range of frequencies is sufficient. The energy in periodic signals is concentrated in this range makes the spectrum available for analysis. This allows today's profusion of communication systems.

Chapter Sections

- 12-1 Overview of Fourier Analysis
- 12-2 Fourier Coefficients
- 12-3 Waveform Symmetries
- 12-4 Circuit Analysis Using the Fourier Series
- 12-5 RMS Value and Average Power
- 12-6 Fourier Transforms
- 12-7 Circuit Analysis Using Fourier Transforms
- 12-8 Impulse Response and Convolution
- 12-9 Parseval's Theorem

- 11-6 Find the driving-point impedance seen by the voltage source in Figure P11-6 and the voltage transfer function $T_V(s) = V_2(s)/V_1(s)$. Insert a follower at point A and repeat.

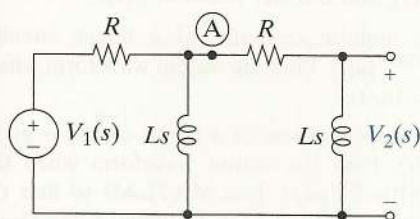


FIGURE P11-6

- 11-7 Find the voltage transfer function $T_V(s) = V_2(s)/V_1(s)$ for Figure P11-7. Select values of R and C so that $T_V(s)$ has a pole at $s = -1$ krad/s.

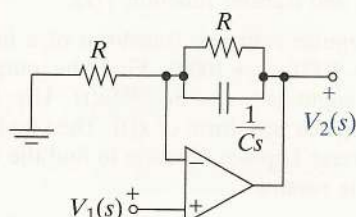


FIGURE P11-7

- 11-8 Find the current transfer function $T_I(s) = I_2(s)/I_1(s)$ in Figure P11-8. Select values of R and L so that $T_I(s)$ has a pole at $s = -500$ rad/s.

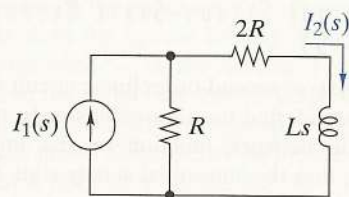


FIGURE P11-8

- 11-9 Find the voltage transfer function $T_V(s) = V_2(s)/V_1(s)$ of the cascade connection in Figure P11-9. Locate the poles and zeros of the transfer function.

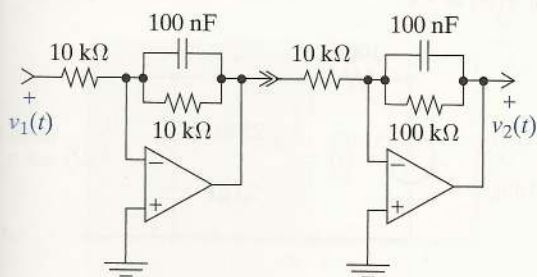


FIGURE P11-9

- 11-10 Find the voltage transfer function $T_V(s) = V_2(s)/V_1(s)$ of the cascade connection in Figure P11-10. Locate the poles and zeros of the transfer function.

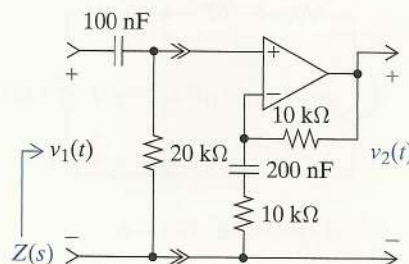


FIGURE P11-10

- 11-11 Find the input impedance $Z(s)$ in Figure P11-10.

OBJECTIVE 11-2 NETWORK FUNCTIONS, IMPULSE RESPONSE, AND STEP RESPONSE (SECTS. 11-3, 11-4)

- Given a first- or second-order linear circuit, find its impulse or step response.
- Given the impulse or step response of a linear circuit, find the network functions.
- Given the impulse or step response of a linear circuit, find the response due to other inputs.

See Examples 11-8, 11-9, 11-10, 11-11, 11-12 and Exercises 11-8, 11-9, 11-10, 11-11, 11-12, 11-13

- 11-12 Find $v_2(t)$ in Figure P11-12 when $v_1(t) = \delta(t)$. Repeat for $v_1(t) = u(t)$.

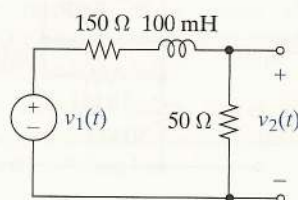


FIGURE P11-12

- 11-13 Find $v_2(t)$ in Figure P11-13 when $v_1(t) = \delta(t)$. Repeat for $v_1(t) = u(t)$.

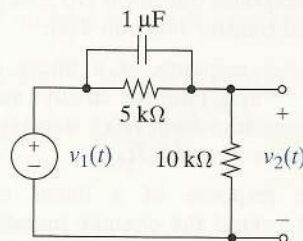


FIGURE P11-13